



DUDEKULA SHAIKSHAVALI

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Google Scholar : <https://scholar.google.com/citations?user=hOaphsYAAAAJ&hl=en>

Date of birth: 30/06/1992 **Nationality:** Indian

ABOUT ME

Motivated academician with a Ph.D. in VLSI domain and experience in teaching, research, and advanced semiconductor device designs. Seeking an Assistant Professor position where I can leverage my expertise in semiconductor physics, VLSI, and microwave engineering to deliver quality education, mentor students, and contribute to cutting-edge research and academic growth.

EDUCATION AND TRAINING

- [2016 – 2023]

Ph.D
Vellore Institute of Technology
City: Vellore | **Country:** India | **Field(s) of study:** VLSI | TCAD Silvaco | Semiconductor Device Physics | **Thesis:** HIGH PERFORMANCE SCHOTTKY BARRIER DIODES FOR MICROWAVE DETECTION
- [2013 – 2014]

M.Tech
SRM University
City: CHENNAI | **Country:** India | **Field(s) of study:** Embedded System Technology | **Final grade:** 84.6 | **Thesis:** BATTERY CHARGING OPTIMIZATION SOLAR POWERED ROBOT
- [2009 – 2013]

B.Tech
G.PullaReddy Engineering College
Address: 518007, KURNOOL, India | **Field(s) of study:** Electronics and Communication Engineering | **Final grade:** 60.5 | **Thesis:** IMAGE RETRIEVAL USING BOTH COLOR AND TEXTURE

WORK EXPERIENCE

- QIS College of Engineering and Technology**
City: Ongole | **Country:** India
- [11/2024 – Current]

Assistant Professor
QIS College of Engineering and Technology
City: Ongole | **Country:** India
- [08/2023 – 05/2024]

Assistant Professor
Responsibilities
 - R&D Coordinator
 - Internship Coordinator
 - BOS member
 - IEEE Robotics and Automation Society Advisor**Courses Assisted**
 - Semiconductor Device Physics
 - Electronics Devices and Circuits

- Electromagnetic waves and Transmissions Lines
- VLSI Design

Vellore Institute of Technology

City: Vellore | Country: India

[2016 – 2019] Teaching cum Research Assistant

PUBLICATIONS

- [2026] **Design and Optimization of AlGa_N/AlN/GaN L-SBD for Radiofrequency Applications**
Authors: Shaikshavali Dudekula, Nilesh Kumar Jaiswal, Thomas Ebel, Venkata Prasad Chowdam, Sang-Mo Koo, Kannadassan Dhanaraj | **Journal Name:** ACS Omega [SCIE; IF: 4.3, Q1] | **Volume, Issue and Pages:** 11(2) | **Publisher:** ACS
- [2025] **Dual-Functional Ni/4H-SiC Schottky Interfaces: Robust UV and X-ray Detection, Imaging in Extreme Environments**
Authors: Chowdam Venkata Prasad*Sunjae Kim....Dudekula Shaikshavali | **Journal Name:** ACS Applied Electronic Materials [SCIE; IF: 4.7, Q1] | **Volume, Issue and Pages:** 8(1) | **Publisher:** ACS
- [2025] **Electrical Performance and Deep-Level Trap Characterization of p-CuGaO₂/β-Ga₂O₃ Heterojunctions for Power Electronics**
Authors: Chowdam Venkata Prasad*Geon-Hee LeeJang Hyeok Park, Dudekula Shaikshavali | **Journal Name:** ACS Applied Materials & Interfaces [SCIE; IF: 8.3, Q1] | **Volume, Issue and Pages:** Vol. 17, Issue 29, | **Publisher:** ACS Publications
- [2022] **Design of microwave β-(Al_xGa_{1-x})₂O₃/Ga₂O₃ lateral Schottky barrier diodes**
Reference:
Authors: Dudekula Shaikshavali and D. Kannadassan | **Journal Name:** Semicond. Sci. Technol, [SCIE; IF: 2.053, Q2] | **Volume, Issue and Pages:** 37(1) | **Publisher:** IOP
- [2021] **Influence of surface trap states on RF/microwave performance of lateral AlGa_N/Ga_N Schottky barrier diode**
Authors: Dudekula Shaikshavali and D. Kannadassan | **Journal Name:** Electromagnetic Waves and Applications [SCIE; IF: 1.3, Q2] | **Volume, Issue and Pages:** 36(1) | **Publisher:** Taylor&Francis
- [2018] **Modeling of Field Dependent Maxwell-Wagner Interfacial Capacitance for Bilayer Metal- Insulator-Metal (MIM) Capacitors**
Authors: Kannadassan, D; Rajshekar, Kadiyam; Shaikshavali, Dudekula; Sanal, Aparna; Baghini, Maryam Shojaei ; Mallick, P. S, | **Journal Name:** Advanced Science Letters (Scopus) | **Volume, Issue and Pages:** 24(8) | **Publisher:** American Scientific Publishers
- [2015] **Battery charging optimization solar powered robot**
Authors: Dudekula Shaikshavali and Ruhab Bevi A | **Journal Name:** International Journal of Applied Engineering and Research (Scopus) | **Volume, Issue and Pages:** 10(6) | **Publisher:** Research India Publications

CONFERENCES AND SEMINARS

- [16/04/2026 – 18/04/2026] **Multi-Stage ML Ensemble Framework for Predicting Organizational KPI Deviation and Policy-Driven Optimization**
 Chittagong, Bangladesh.
 Published in IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)

- [26/11/2025] **Ravuri Vivekananda chowdary, Dudekula Shaikshavali, "Design of Power Optimized 1bit ALU Using Multiplexers in Cadence system 90 nm Technology"**
Ongole, India.
Published in Proceeding of IMCEST-2025 International Multi-Conference on Engineering, Science & Technology.
- [06/11/2025 – 07/11/2025] **D Shaiksha Vali; Ala'a Al-Shaikh; Venkata Ramaiah Kavuri; SkGouse John; Sravanthi; R Suresh Kumar, "Low-Power VLSI Accelerator Architecture for AI-Enhanced Real-Time Audio and Video Processing at the Network Edge"**
Chennai, India.
- [06/11/2025 – 07/11/2025] **R Sabitha; D Shaikshavali; M Prathyusha; Riham Alkabbji; J Sunil Kumar; E D Kanmani Ruby, "An Enhanced Glaucoma Disease Detection through Deep CNN Using Fundus Data with Self-Attention-based Grey Wolf Optimization"**
Chennai, India.
Published in Tenth International Conference on Science Technology Engineering and Mathematics (ICONSTEM)
- [11/02/2022 – 13/02/2022] **K. Rajshekhar, Shaikshavali D, Kannadassan D, "Effect of Al-doping in p-type SnOx Thin Film Transistors: Physics and Modeling"**
New Delhi, India.
Published in IEEE conference proceedings(DELCON-2022)

PROJECTS

- [02/01/2025 – Current] **Smart Sanitary Pad System for Caregiver-Assisted Menstrual Hygiene**
Caregivers of mentally challenged and disabled girls and women face significant difficulties in managing menstrual hygiene. Many individuals are unable to recognize or communicate the need to change their sanitary pads, which can lead to leakage, discomfort, and serious hygiene-related health risks such as infections and skin irritation. To address these challenges, the project team developed an **Innovative Smart Sanitary Pad System** that leverages sensor-based monitoring and real-time alerts. A grant of **Rs. 43,500** is awarded for prototype from IEEE WIE section.

BOOK CHAPTERS

- [25/03/2024] **Innovative Technologies and Intelligent Systems: Transforming the Future, ISBN-13 (24) 9788119821006, 2024 Royal Book Publishing**

PATENTS

INTELLIGENT SANITARY PAD WITH SMART MOISTURE SENSOR AND NOTIFICATION

Design number: DE 20 2025 106 199 U1

Published: 04 November 2025

Registration date: 11 February 2025

Type: Utility

Country: Germany

AI BASED SPECTACLES FOR VISUALLY IMPAIRED

Design number: 475595-001

Granted: 16 January 2026

Registration date: 01 October 2025

Type: Design

Country: India

AI BASED SYSTEM AND METHOD FOR MEAL RECOMMENDATION

Design number: 202541017894

Published: 14 March 2025

Registration date: 28 February 2025

Type: Utility

Country: India

SOLAR POWERED AGRICULTURE ROBOT

Design number: 6347636

Grant date: 27 February 2024

Registration date: 19 February 2024

Type: Design

Country: UK

WORKSHOPS AND COURSES

Workshops

- Organised drone workshop for 2 days i. e from 16-04-2026 to 17-04-2026 in QISCET, Ongole.
- Attended "The Future of VLSI Design for Next-Generation Communication Standards", by Communication Society, IEEE Hyderabad Section, May, 2024.
- Attended "Online TCAD –Circuit Simulation Workshop" dually organized by IIT Bombay and Synopsys during 1-5 August 2022.
- Participated in "World Nano Congress on Advanced Science & Technology" (WNCST-2021) organized by Centre for Nanotechnology Research (CNR) during 8-13 March 2021.
- Attended "Frontiers of Excellence in Wide and Ultrawide Bandgap Semiconductors and Electronic Systems" workshop to be held during December 14-15, 2019 at the IIT Bombay, India.
- Attended "INUP Hands-on Training Workshop on Nanofabrication Technologies" to be held at IIT Bombay during 22-05-2017 to 24-05-2017.

Courses

1. Short course program on "Modeling and Simulation of Nano-transistors" at IIT, Kanpur from 13-17th February 2017.
2. Short course program on "Flexible Electronics 2017" at IIT, Kanpur from 3-8th July 2017.

LANGUAGE SKILLS

Mother tongue(s): Telugu **Other language(s):** English, Hindi and Tamil.

TECHNICAL KNOWLEDGE

Software Tools

- TCAD (Silvaco and Sentaurus) , AWR Microwave office
- Cadence, MATLAB, Xilinx ISE

ORGANISATIONAL SKILLS

Organisational skills

- Editor and Organiser of a 2 day National level Symposium "**NATIONAL-LEVEL STUDENT SYMPOSIUM ON PIONEERING IDEAS, ANALYSIS, RESEARCH & KNOWLEDGE- S.P.A.R.K-2025**", Oct 24-25, 2025, QISCET, Ongole.
- Member of organizing committee of QISFEST, Feb 6-8, 2024, 2025, &2026 at QISCET, Ongole,
- Member of organizing committee of 2 International Conference on Nanoscience and Nanotechnology2019 (ICNAN'19), Vellore Institute of Technology, India.
- Member of organizing committee of 1 International Conference on Nanoscience andNanotechnology2016 (ICNAN'16), Vellore Institute of Technology, India.
- Member of "CAD (Center for Awareness and Discussion)" & "VSC (Vivekananda Study Circle)"from 2011-2012, G.Pulla Reddy Engineering College, India.

PROFESSIONAL MEMBER-SHIPS

IEEE Member

IEEE Robotics and Automation Society Member and Advisor

NPTEL CERTIFICATION

[21/07/2025 – 12/10/2025]

Semiconductor Devices and Circuits

FDPS

Pedagogical Strategies for 21st Century Learners by National Institute of Technical Teachers Training and Research

Location: QISCET Ongole. Date: 23/06/2025 to 27/06 /2025

Familiarisation Workshop on Quantum Computing

Location: IISC Bangalore. Date: 22/09/2025 to 26/09/2025

Semiconductor Devices and Circuits

Location: NPTEL Date: 12/10/2025

Advances in Semiconductor Technology for Next-Generation Devices

Location: IIITDM Kurnool. Date: 01/12/2025 to 06/12/2025

Emerging Technologies in Wireless Communications

Location: NEC, Palnadu. Date: 14/07/2025 to 19/07/2025

REFERENCES

Dr. Ch. Hima Bindu

Principal, Bapatla Womens Engineering College, AP.

Mobile: +91 9885396299

I have know him for his dedicated work. I strongly recommend.

Dr. D. Kannadassan

Professor & HoD, SENSE, Vellore Institute of Technology, Vellore, India

Mobile: 9944055243

I strongly recommend Dr. Dudekula Shaikshavali. Any further information, please do not hesitate to contact me.